Missing Data Visualization

Workshop 4 — Module 4

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Data Visualization for Data Scientists
UNYT Tirana

Module Roadmap

1. Why missing data matters for visualization
2. Three missing data mechanisms: MCAR, MAR, MNAR
3. The missingness matrix and % bar chart
4. Shadow-augmented plots: making the invisible visible
5. Co-occurrence patterns: UpSet plots and dendrograms
6. Visualizing imputation effects

Learning Outcome
You will diagnose missingness patterns before plotting, determine whether data is MCAR/MAR/MNAR, and use shadow-augmented plots to reveal what standard charts silently discard.

Why Missing Data Matters

The Silent Problem

Most visualization tools **silently drop** missing rows:

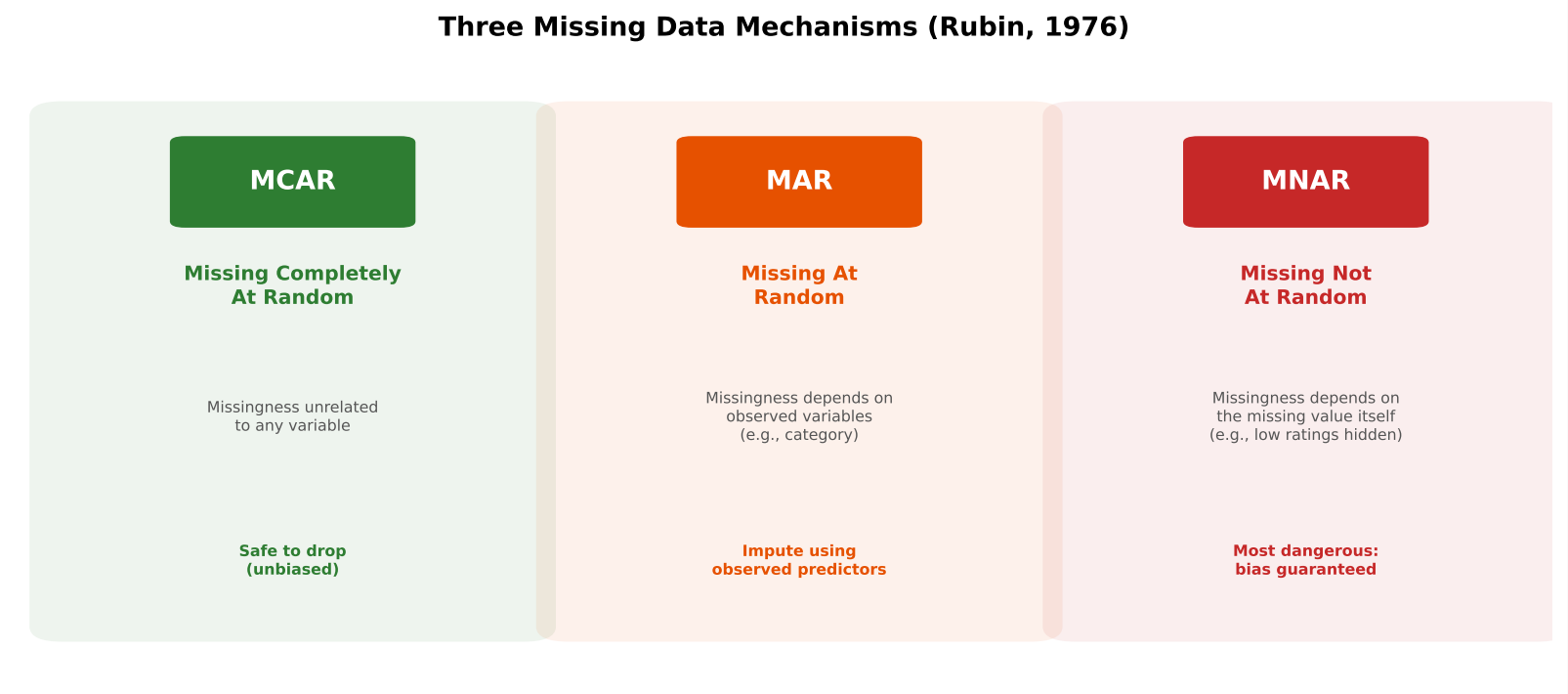
- `ggplot2`: "Removed 847 rows containing missing values"
- `matplotlib`: NaN values produce gaps or are skipped
- `seaborn`: drops NA before computing statistics

If 15% of your data is missing and it is **not random**, every chart you produce is biased — but you will never know unless you explicitly visualize the missingness itself.

Pro-Tip
Rule: **before any EDA plot, first plot the missingness**. This is Step 0 of any analysis. If you skip it, you are making invisible assumptions about which data you are not seeing.

Three Mechanisms

MCAR vs MAR vs MNAR (Rubin, 1976)



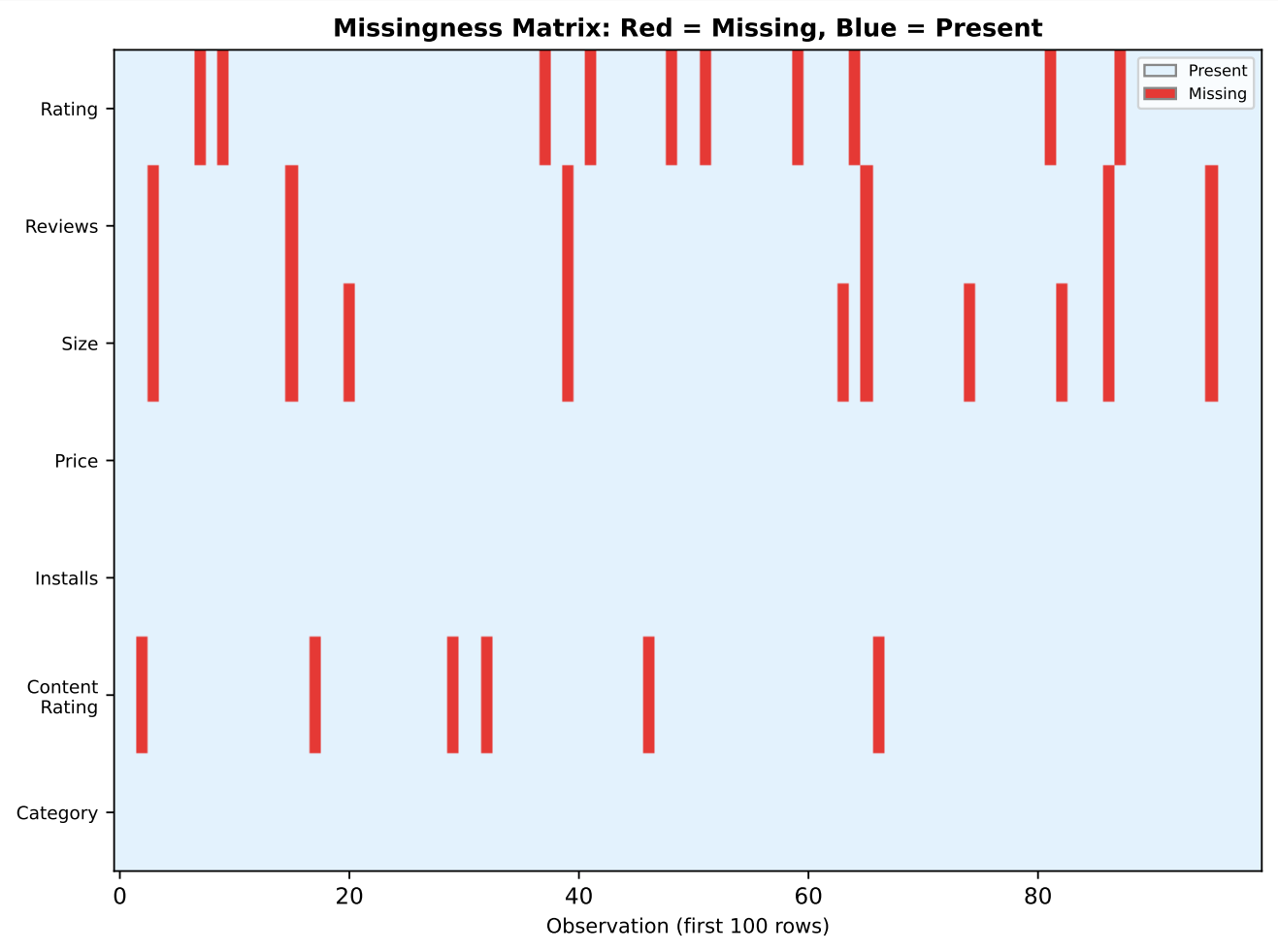
How to Diagnose the Mechanism

Mechanism	Diagnostic	Tool
MCAR	Compare distributions of observed data between "has NA" vs "no NA" groups. No difference ⇒ MCAR.	Little's test, t-tests
MAR	Missingness in column A correlates with values in column B. A depends on observed B.	Missingness dendrogran
MNAR	Can't fully test — but if paid apps hide ratings more than free apps, and you can't predict it from other columns, suspect MNAR.	Domain knowledge, sen

Pro-Tip
MNAR is the most dangerous: the missing values are systematically different from observed values. No imputation method fully fixes MNAR bias — you must acknowledge the limitation.

Visualizing Missingness

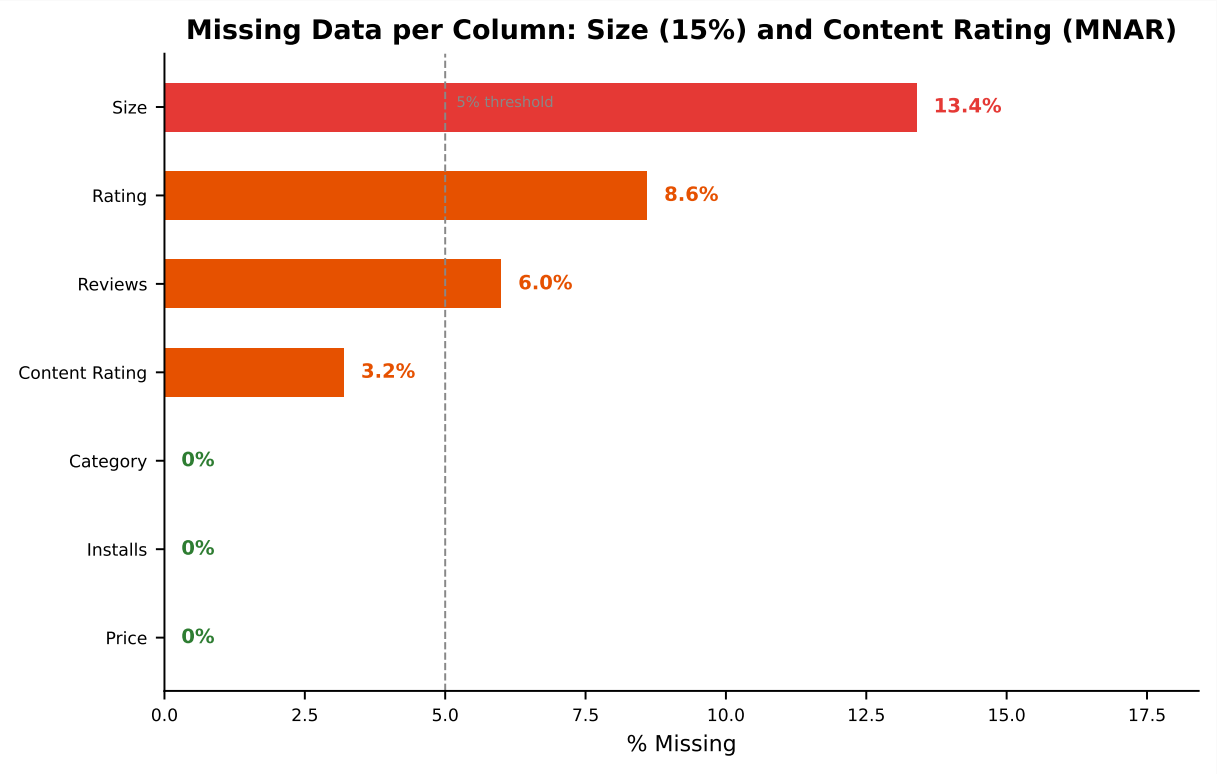
The Missingness Matrix



Data Insight

Each row is an observation; each column is a variable. Red cells are missing. **Vertical stripes** = one column heavily missing. **Horizontal bands** = certain rows missing everything. **Block patterns** = structured

Missing % per Column



Missingness Matrix & Bar in R and Python

R (naniar + visdat)

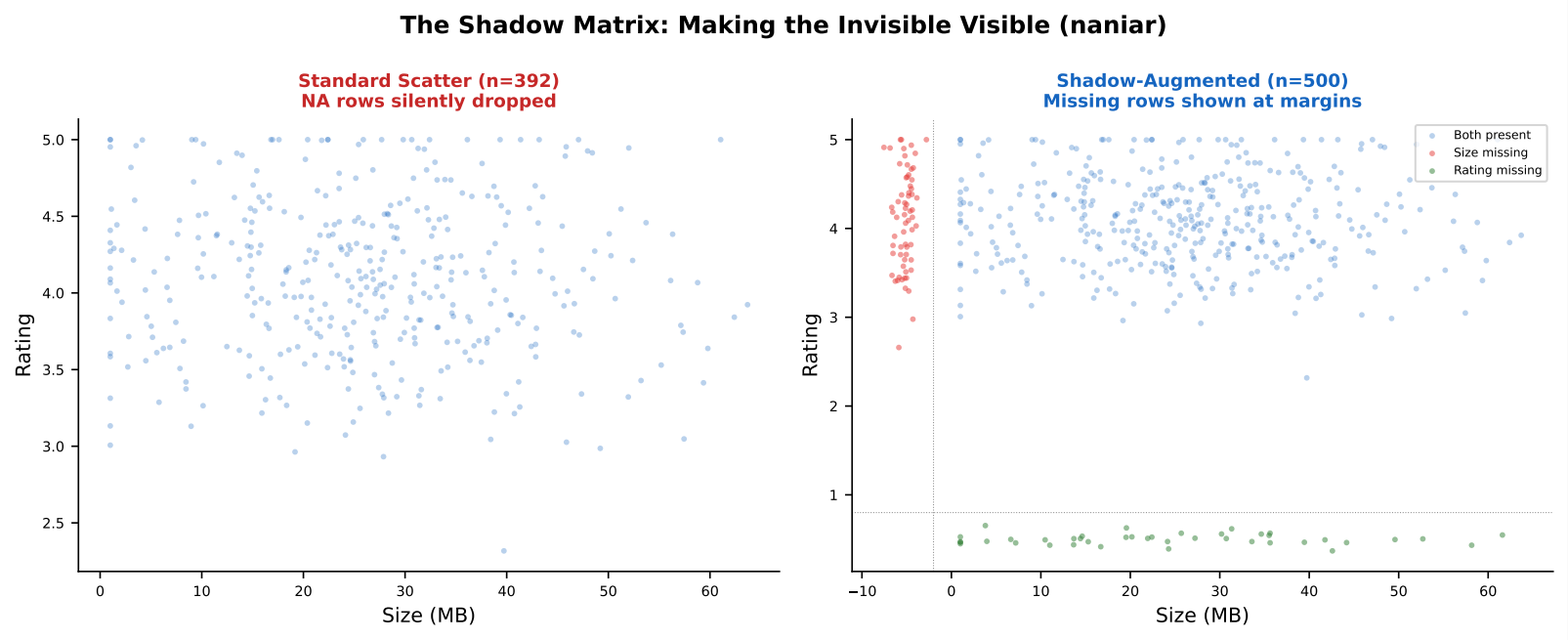
```r
library(naniar); library(visdat)

# Full type + NA matrix
vis_dat(apps)      # colour by
    type
vis_miss(apps)     # colour by NA

# Missing % per variable
gg_miss_var(apps) # horizontal
    bar
gg_miss_var(apps, facet = Type)

# Summary table
miss_var_summary(apps)
#>   variable    n_miss pct_miss
#>   Size            75     15.0
#>   Rating          40      8.0
#>   ...

# Little's MCAR test
mcar_test(apps |> select(
  Rating, Size, Reviews))
```

Python (missingno + pandas)

```python
import missingno as msno

# Missingness matrix
msno.matrix(apps, figsize=(8,5))

# Missing % bar
msno.bar(apps, figsize=(8,4))

# Built-in pandas
(apps.isna().mean()
    .sort_values(ascending=False)
    .plot.barh(color="#E53935"))

# Summary
apps.isna().sum()
apps.isna().mean() * 100

# Heatmap of nullity correlation
msno.heatmap(apps, figsize=(6,5))
```

Shadow-Augmented Plots

Making the Invisible Visible



The Shadow Matrix (naniar)

R (naniar)

```r
# geom_miss_point: shows NA rows
# at the margin of a scatter plot
ggplot(apps, aes(x = Size,
                 y = Rating)) +
  geom_miss_point(alpha = 0.3) +
  theme_minimal() +
  labs(title = "Shadow Scatter")

# bind_shadow: add _NA columns
apps_shadow <- apps |>
  bind_shadow()  # adds Rating_NA
       etc.

# Then facet by missingness
ggplot(apps_shadow, aes(
    x = Reviews, fill = Rating_NA
        )) +
  geom_histogram(bins = 30,
    position = "identity",
    alpha = 0.5) +
  scale_fill_manual(values =
    c("!NA" = "#1565C0",
      "NA"  = "#E53935"))
```

Concept

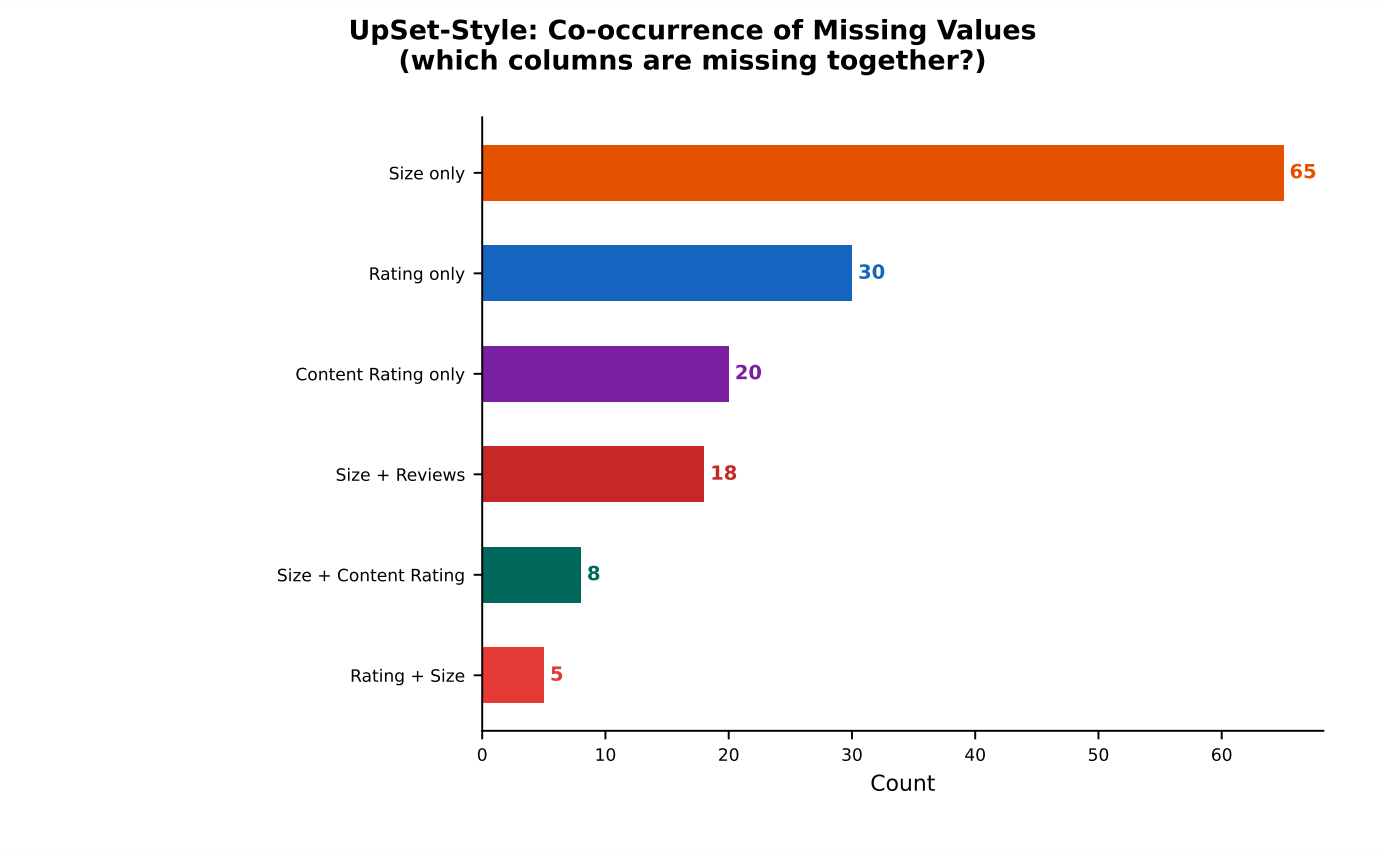
The **shadow matrix** creates a parallel Boolean DataFrame where each cell records "is this value missing?"

`bind_shadow()` (R) or `df.isna().add_suffix("_NA")` (Python) appends these columns, enabling:

- Colour by missingness in scatter plots
- Facet histograms by "missing vs present"
- Test whether the distribution of X differs between "Y missing" vs "Y present"

Co-occurrence Patterns

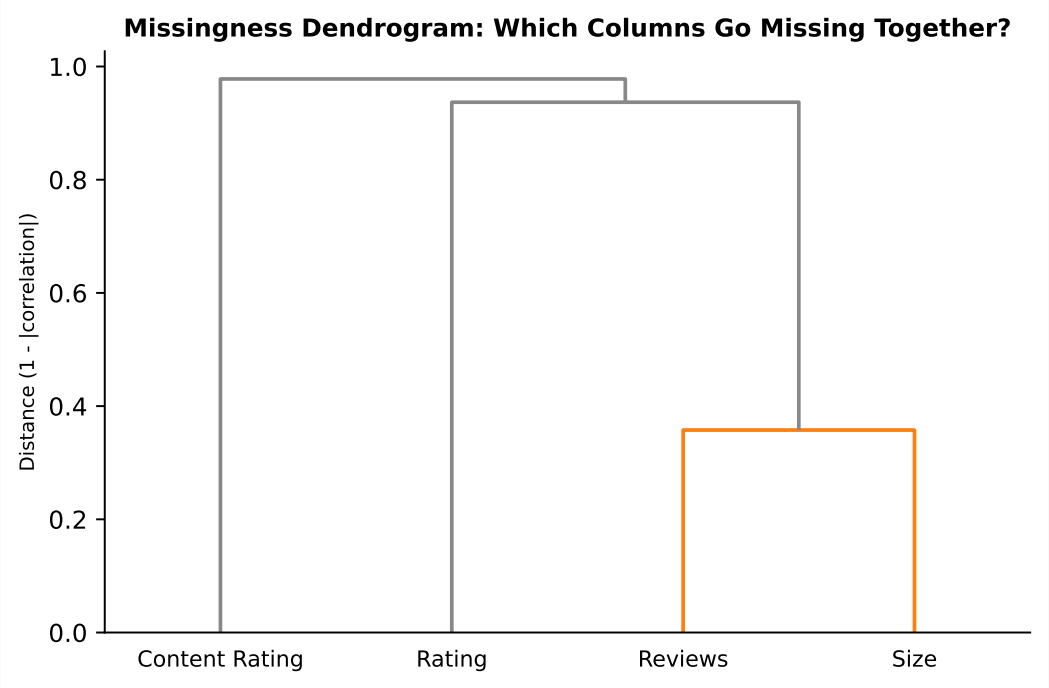
UpSet Plot: Which Columns Go Missing Together?



Data Insight

The UpSet pattern reveals **cascading missingness**: when Size is missing, Reviews is often missing too (18 co-occurrences). This suggests a shared data-entry failure, not independent missingness. In R: `gg_miss_upset(apps)`. In Python: `missingno` or `upsetplot` package.

Missingness Dendrogram

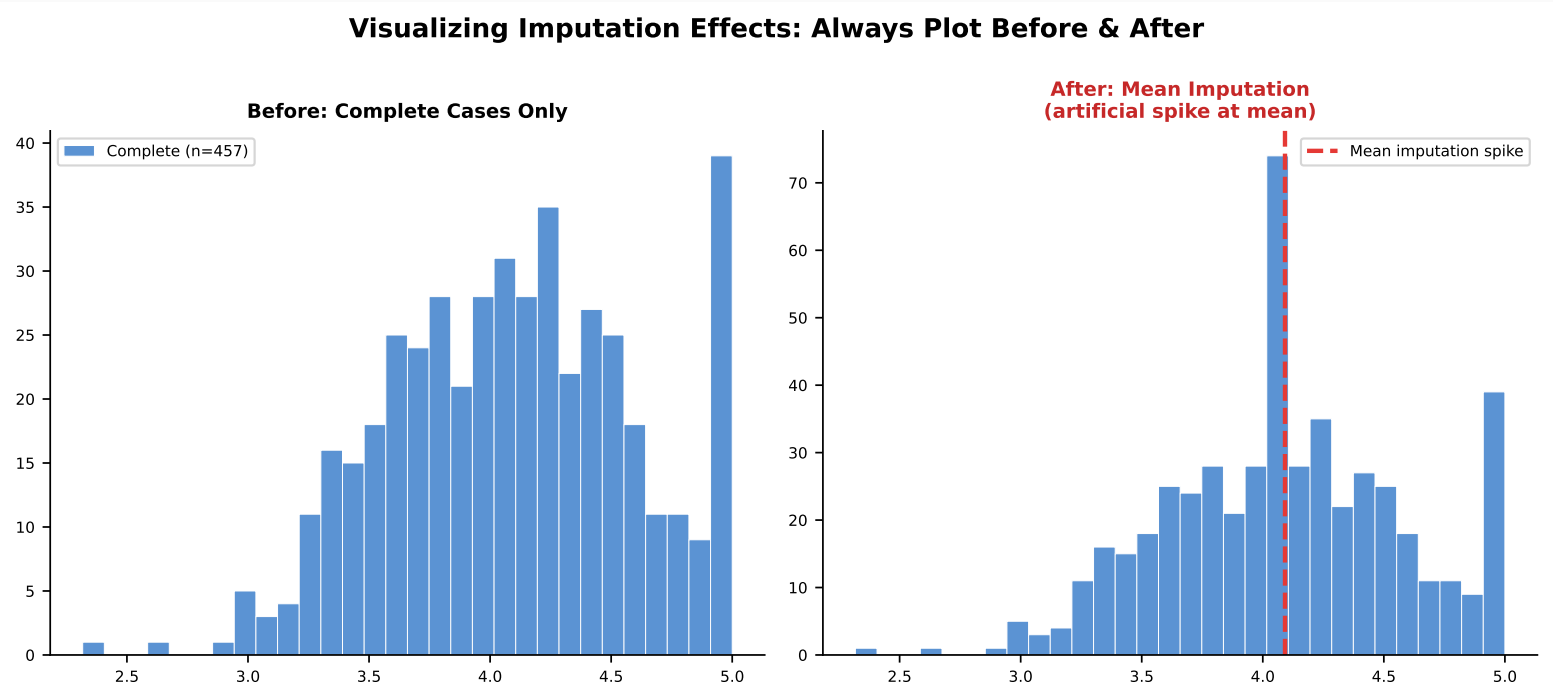


Pro-Tip

The dendrogram clusters columns by the **correlation of their missingness indicators**. Columns that cluster together tend to go missing in the same rows. In R: `naniar` or manual `hclust()`. In Python: `msno.dendrogram()`.

Imputation Effects

Visualizing Before & After Imputation



Imputation Strategies at a Glance

Method	Effect on Distribution	When OK
Drop rows	Reduces n; biased if not MCAR	MCAR, few NAs
Mean/median	Spike at centre; shrinks variance	Never ideal
Mode (categorical)	Inflates modal category	Rare categories
Regression	Preserves trend; underestimates variance	MAR
Multiple (mice/MICE)	Multiple plausible datasets; correct CI	MAR, gold standard

Pro-Tip

Always **plot the distribution before and after imputation** (overlay histograms, Q-Q plots). If the imputed distribution looks different from the observed, document the discrepancy. In R: `mice` package. In Python: `sklearn.impute` or `miceforest`.

Package Ecosystem

Missing Data Visualization Packages

Missing Data Visualization Packages

Package	Lang	Key Functions	Notes
`naniar`	R	`Shadow matrix, gg_miss_var, gg_miss_upset, geom_miss_point`	Most comprehensive; Tierney & Cook (2023)
`visdat`	R	`vis_dat(), vis_miss()` `colour-coded type + NA matrix`	Quick first look; Tierney (2017)
`missingno`	Python	`msno.matrix(), msno.bar(), msno.heatmap(), msno.dendrogram()`	Biswas (2016); matplotlib-based
`pandas`	Python	`df.isna().sum(), df.info(), df.isna().mean().plot.barh()`	Built-in; no install required

EDA Checklist

Missing Data EDA Checklist

Step	Action	Code
1	How much is missing?	`gg_miss_var()` / `msno.bar()`
2	Where is it missing?	`vis_miss()` / `msno.matrix()`
3	Which columns co-occur?	`gg_miss_upset()` `msno.dendrogram()`
4	Is it MCAR/MAR/MNAR?	Shadow histograms + domain kn
5	How does imputation affect shape?	Overlay before/after histograms
6	Document and proceed	Report % missing, mechanism, st

Synthesis

Key Takeaways

1. **Plot the missingness before any analysis**. It is Step 0. Standard charts silently drop NA rows, hiding potential bias.
2. **Three mechanisms** (Rubin, 1976): MCAR (safe to drop), MAR (impute from observed), MNAR (bias guaranteed).
3. The **missingness matrix** reveals vertical stripes (one bad column), horizontal bands (bad rows), and block patterns (structured missingness).
4. **Shadow-augmented plots** colour or facet by missingness, revealing whether missing rows differ from observed rows.
5. **UpSet plots** and **dendrograms** identify co-occurring missingness (cascading data-entry failures).
6. **Always visualize imputation effects**: overlay before/after distributions to detect artificial spikes.

Essential Readings

- Rubin, D. B. (1976). "Inference and Missing Data." *Biometrika*, 63(3), 581–592.
- Tierney, N. J. & Cook, D. (2023). "Expanding Tidy Data Principles to Facilitate Missing Data Exploration." *Journal of Statistical Software*, 105(7).
- van Buuren, S. (2018). *Flexible Imputation of Missing Data*, 2nd ed. CRC Press.
- Biswas, A. (2016). missingno: Missing Data Visualization for Python. `https://github.com/ResidentMario/missingno`

Next Module: Outlier Detection & Visual Diagnostics (W04-M05)